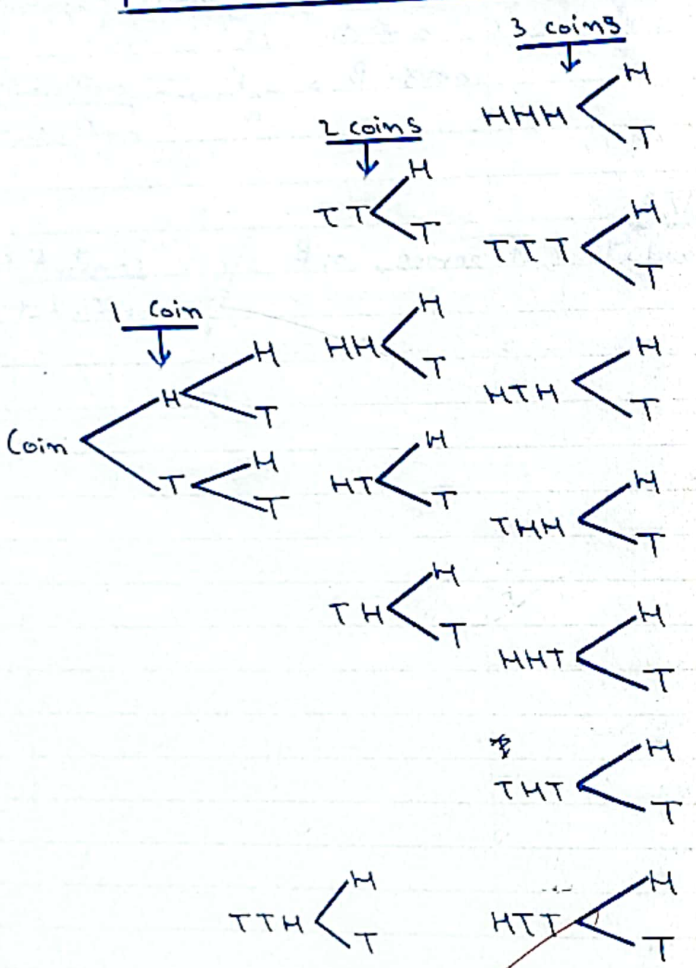
Formula of Probability

$$P(E) = \frac{\text{No. of favourable outcomes}}{\text{Total no. of outcomes}}$$



Teacher's Signature: .....

Experiment No. 3

Dated : .....

Objective : To determine the probability of (a) Case of cards (b) Cases of cards (c) Double dice (d) Cases of 3 coins

Case A:

Tossing of 3 Coins

Material Required:

* 3 coins

Formula of Probability:

$$P(E) = \frac{\text{No. of favourable outcomes}}{\text{Total no. of outcomes}}$$

Pre-Requisite Knowledge:

* A coin has 2 faces which are 'Head' and 'Tail'.
* Probability of getting head is $P(H) = \frac{1}{2}$

Observation:

* The No. of cases increases as we increase the number of dies.

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Teacher's Signature: .....

Experiment No. .....

Dated : .....

Objective : .....

.....

Result:

i) Total no. of outcomes = 8

ii) Find the probability when all the faces are same?

No. of favourable outcomes = 2 (HHH, TTT)

$$P(E) = \frac{2}{8} = \frac{1}{4}$$

iii) Find the probability of getting atleast two heads?

No. of favourable outcomes = 4 (HHH, HHT, HTH, THH)

$$P(E_2) = \frac{4}{8} = \frac{2}{4} = \frac{1}{2}$$

Case B:

Cases of Double Dies

Pre-Requisite Knowledge:

i) Rolling 1 Die

Total no. of outcomes = 6

ii) Rolling 2 Dies

Total no. of outcomes = $6 \times 6 = 36$

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Sample space ->

	1	2	3	4	5	6
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

Teacher's Signature: .....

Experiment No. .....

Objective : .....

Dated : .....

Observation:

The no. of outcomes increases as the no. of dices increases

Result:

(i) Find probability of getting non-duplets.

→ No. of favourable outcomes = 36 − 6 = 30

$$P(E) = \frac{30}{36} = \frac{5}{6}$$

(ii) Find probability when product of no. on both faces is more that and equal to 36. Is it sure event.

→ No. of favourable outcomes = 36 − 1 = 35

$$P(E) = \frac{35}{36}$$

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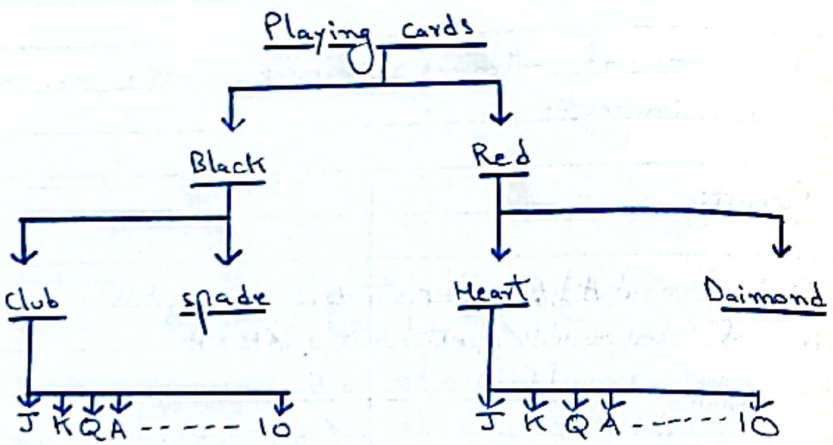

Teacher's Signature: .....

Experiment No. .....

Objective : .....

Dated : .....

Case - C

Cases of Playing cards

Pre-Requisite Knowledge:

Total No. of outcomes = 52

No. of face cards = 12

No. of suits = 4 (♢, ♡, ♧, ♤)

Observation:

No. of Non-face cards = 52-12 = 40

Result:

Case → All daimonds and clubs cards removed from the pack.

Total no. of outcomes = 52 - 26 = 26

i) No. of non-face cards

$26 - 8 = \frac{18}{26}$ $\quad P(E) = \frac{18}{26}$

ii) No. of face cards

$26 - 18 = 8$ $\quad P(E) = \frac{8}{26}$